

Exhaustive AI-assisted review of SPM: eighty-three verified defects, five fixes sent upstream, two merged and measured on SPM's own data

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Abstract

SPM is among the most used neuroimaging packages and, like its peers, is large, old, mathematically dense, and read far less often than it is run. I asked an AI coding agent to read all of it adversarially, subsystem by subsystem, and to report nothing it could not reproduce by executing code. From the development source at commit 530ec52 (23 August 2026), divided into eleven subsystems, the review yielded 83 confirmed defects, each with file-and-line evidence and a numerical reproduction, and about 38 lower-confidence candidates recorded separately. The most reassuring result frames the rest: the standard fMRI general linear model, from `spm_spm` through the T and F Euler-characteristic densities and both false-discovery procedures, was traced end to end and held. The defects sit in the tiers around that path. Five were ranked high priority by severity times reach and sent upstream as ordinary pull requests with issues carrying runnable reproductions: the χ^2 Euler-characteristic densities in `spm_ECDensity` use the wrong power of the threshold at dimensions two and three, so peak inference on χ^2 maps is far too conservative and cluster inference far too liberal (a new unit test fails 12 of 23 assertions on the shipped code and none after the fix); the variational free energy in `spm_nlsi_GN` counts its log-determinant term n_q times too often, which reaches DCM for evoked responses; `@meeg/badsamples` adds the peristimulus baseline twice when mapping artefact events to samples; `spm_eeg_downsample` stamps the requested rather than the achieved sampling rate; and `spm_design_contrasts` pads parametric-modulation columns without the basis-function factor. Within two days the maintainers merged the two M/EEG fixes; the other three are under review. To measure what the merged fixes mean in practice I ran SPM's own M/EEG code, executed in GNU Octave, on SPM's mismatch-negativity tutorial recording. Before the fix, every artefact window on epoched data sat 100 ms late, so the detector excluded only 24% of the artefact it had found and 71% of what it excluded was clean; yet the final robust-averaged mismatch negativity moved by only 1.5% RMS, because robust weighting cushions the mask. The sampling-rate defect, which fires only for the explicit `decimate` and `downsample` methods, reported a 915 s recording as 781 s and a 275 ms mismatch-negativity peak at 235 ms. Both reach corrections narrow the findings and are reported as such. The reviews, harnesses, patches, pipeline and results are public.

1 Introduction

Statistical Parametric Mapping [1] has carried a large share of published functional neuroimaging for three decades, and its M/EEG and dynamic causal modelling (DCM) branches carry much of the electrophysiology and effective-connectivity literature besides. The code base is excellent and heavily used, and both facts work against complete review: a routine that is correct for the inputs its author had in mind can be wrong for inputs

that became common later, and a test suite cannot cover a regime nobody thought to test. When such a defect surfaces, as with cluster inference [2], it is treated as a singular event rather than as one instance of a class that could be searched for.

A companion paper reports the same method applied to AFNI, together with an end-to-end reanalysis of one merged fix on forty participants of an open dataset [3]. This paper is the SPM half, kept separate so that each package’s findings can be read on their own terms. Its design has two deliberate features. The code review came first and any literature search second, so that findings were not filtered by what I expected to matter. And nothing was reported to the maintainers without a numerical reproduction and, where possible, a patch, so that the unit of output is a pull request rather than a suspicion. Section 3 reports what was found, what was filed and what the maintainers merged. Section 4 then takes the two merged fixes and asks what they change on SPM’s own tutorial data, using SPM’s own code.

2 Methods

2.1 Exhaustive adversarial review

An AI coding agent (Claude, Anthropic) worked in a sandboxed environment with the SPM source checked out (spm/spm development version at 530ec52, 23 August 2026), a C compiler, Python with NumPy and SciPy, and later GNU Octave. The source was divided into eleven subsystems, each read in full by a separate agent with an adversarial brief: for every routine, what inputs are legal but unanticipated, what happens at boundaries and at zero, where do numeric types and index bases change, where is a convention assumed rather than checked. The subsystems were the statistical distribution library; random field theory and the results machinery; the GLM estimation core including ReML; spatial preprocessing; image and file I/O; the C and MEX sources; DCM and Bayesian inversion; numeric and mesh utilities; M/EEG processing including the @meeg object; the spatial toolboxes (FieldMap, DARTEL, Shoot, Longitudinal, OldNorm, OldSeg, Spatial); and the active-inference code (spm_MDP_* and its tensor helpers). A human directed the agent at the level of package, subsystem and which candidates to pursue; the agent wrote the reviews, harnesses and patches. Every candidate was recorded with file and line evidence, and code paths that were checked and found correct were recorded with the same care.

2.2 Verification by execution

No candidate was reported without execution of one of three kinds: transcribing the routine faithfully to Python, idiosyncrasies included, and testing the transcription against NumPy, SciPy, closed forms or Monte Carlo; checking a claimed identity numerically (for the Euler-characteristic densities, that $\chi_1^2 = Z^2$ and $\chi_\nu^2 = \nu F(\nu, \infty)$, and for the Wishart divergence, a Monte-Carlo estimate of the true Kullback–Leibler value); or running the shipped MATLAB code itself. The third kind became possible once SPM was built for GNU Octave 8.4 with `make PLATFORM=octave`. Two of SPM’s dependencies had to be replaced for that: FieldTrip’s 24-bit BDF reader, which ships only as a MEX binary, was replaced by an equivalent pure-MATLAB decoder verified against MNE-Python’s independent decode to four decimals, and a handful of MATLAB-only string helpers were shimmed. Candidates that did not survive execution were recorded as plausible, not confirmed, and are not counted among the 83.

2.3 Triage and upstream filing

Confirmed findings were ranked by severity times how often the code path runs in real pipelines, with silent failure weighted above a crash, into three tiers. The top tier was patched on a fork as one branch per fix, each with a commit message and, for the first, a new unit test in the repository’s own `matlab.unittest`

pattern. Following SPM’s contributing guide, each fix was filed as a GitHub issue stating expected and actual behaviour with a runnable MATLAB snippet, paired with a pull request. Each pull request states what testing was done and what was not: the χ^2 fix and the two M/EEG fixes were executed in Octave, the other two were verified by transcription and tracing only. The maintainers’ review is the adjudication.

2.4 Literature exposure, assessed afterwards

Exposure was assessed only after the review and only for the top tier, by anchoring each finding to the methods papers that document the affected machinery, rather than by a per-paper join of the kind the AFNI companion performs for one finding. That join for SPM is future work and its absence is a stated limitation.

2.5 Measuring the merged fixes on SPM’s tutorial data

Data. SPM’s mismatch-negativity (MMN) tutorial recording (`eeg_mmn/subject1.bdf`): one participant, 128-channel Biosemi EEG at 512 Hz, 915 s, passive auditory oddball with 480 standards and 120 deviants [6, 7].

Artefact-window defect (@meeg/badsamples, PR 163). SPM’s own “advanced artefact removal” chapter for this dataset was run with the real SPM functions in Octave: downsample to 200 Hz, band-pass 0.5–30 Hz, epoch –100 to 400 ms with baseline correction, artefact detection in `mark` mode (z-scored-difference detector, 100 ms excision window, bad-channel threshold 1), then robust averaging with “remove bad data” on, and MMN as deviant minus standard. The pipeline was run twice, with `badsamples.m` at 530ec52 and at upstream `main` after the merge; everything else was identical. The two bad-sample masks were compared directly, and the two MMN waveforms by global field power.

Sampling-rate defect (`spm_eeg_downsample`, PR 165). The recording was downsampled from 512 to a requested 200 Hz with `method = 'decimate'` using the real `spm_eeg_downsample` in Octave, before and after the fix, and the stamped rate and the reported time of the last sample were read from the resulting object. The consequence for the MMN was computed by a Python transcription of the same arithmetic (decimation by the rounded factor, 0.5–30 Hz, average reference, epoch –100 to 400 ms): the true waveform was placed on the true time axis and on the axis the pre-fix stamp implies. The default `resample` method and the no-toolbox `fft` fallback are exact, so the defect was also confirmed *not* to fire there.

3 Results: what was found and what was merged

3.1 Counts and the path that held

The review produced 83 confirmed defects, each producing wrong numbers, wrong geometry, a crash or silent data corruption on a concrete input, plus about 38 plausible candidates tracked in the component reviews. Triage placed 5 in the high-priority tier, 15 in the medium tier and 63 in the low tier. Against them stand the paths that were checked and held: the Z, T and F Euler-characteristic densities, `spm_P_RF` in both its unified and Poisson-clumping forms, both FDR procedures, Bonferroni, `spm_resels` and `spm_est_smoothness`; the GLM core (`spm_spm`’s whitening and degrees-of-freedom machinery, the Satterthwaite trace identities, the ReML gradients and curvatures, `spm_hrf`, filter construction, microtime onsets); the NIfTI quaternion round-trip to 3×10^{-12} over 20,000 rotations and every 0-based/1-based voxel convention traced; `spm_matrix` and `spm_imatrix` including negative determinants; the DCM hemodynamic equations, Jacobians, `spm_BMS` and the Bayesian model-reduction closed form; the mainstream robust-averaging routine `spm_robust_average`; and the `spm_dot` and `spm_kron` tensor conventions

Table 1: The five high-priority findings, their fixes, and upstream status as of 3 September 2026. Each pull request is paired with an issue carrying a runnable reproduction. “Executed” means the fix was run against the real SPM code in Octave; the others were verified by transcription and tracing, which each pull request states.

	Defect	Verified	Issue / PR	Status
SP1	<code>spm_ECDensity</code> : χ^2 EC densities use $t^{(\nu-1)/2}$ at every order where Worsley’s densities need $t^{(\nu-d)/2}$; 2-D density inflated by $\sqrt{t}$, 3-D by t	executed, unit test	158 / 159	open
SP2	<code>spm_nlsi_GN</code> : free energy applies the n_q factor to an already Kronecker-expanded precision, overcounting the log-determinant n_q -fold; the M-step gradient is that of the corrected term	traced	160 / 161	open
SP3	<code>@meeg/badsamples</code> : peristimulus baseline added twice when mapping artefact events to samples; every window on epoched data shifted by the baseline	executed, real data	162 / 163	merged 2 Sep
SP4	<code>spm_eeg_downsample</code> : achieved rate computed and printed, requested rate stored	executed, real data	164 / 165	merged 2 Sep
SP5	<code>spm_design_contrasts</code> : parametric-modulation padding uses $\sum h$ columns where the design carries $\sum h \cdot n_{\text{bases}}$; automatic factorial contrasts slide onto the wrong regressors when derivatives are on	numerically	167 / 168	open

against `einsum`. A standard fMRI analysis with the default HRF, realign, coregister, normalise, smooth, T and F contrasts, and RFT or FDR thresholding runs on verified code from end to end. The defects cluster in the tiers out from it: specific statistic types, optional features, particular toolboxes, rare formats and edge cases.

3.2 The high-priority tier

Table 1 lists the five findings sent upstream and their status. Each is described briefly here; the component reviews hold the file-and-line evidence.

SP1: χ^2 Euler-characteristic densities. Random field theory gives the expected Euler characteristic of an excursion set as a sum over dimensions of resel counts times EC densities [4]. In `spm_ECDensity` the χ^2 branch computes a single factor $t^{(\nu-1)/2}$ and reuses it at orders two and three, where the densities need $t^{(\nu-2)/2}$ and $t^{(\nu-3)/2}$; the 2-D density is therefore inflated by $\sqrt{t}$ and the 3-D density by t (Fig. 1A). Three independent checks agree: a transcription cross-checked through the identities $\chi_1^2 = Z^2$ and $\chi_\nu^2 = \nu F(\nu, \infty)$; a Monte-Carlo simulation of smooth χ_4^2 fields on a 256^2 torus (300 realisations, FWHM 8 pixels), where the empirical expected Euler characteristic at thresholds 12, 16 and 20 was 59.5, 15.1 and 3.2 against 60.5, 15.8 and 3.5 from the corrected formula and 209.5, 63.1 and 15.6 from the shipped one; and, after patching, execution of the real MATLAB code in Octave, where the new `tests/test_spm_ECDensity.m` fails 12 of 23 assertions on the shipped code and none on the fix, with the untouched Z, T and F branches bit-identical. The function had no test before. End to end, on a whole-brain χ_{10}^2 map at threshold 40, `spm_P_RF` gives a peak family-wise p of 1.000 where the correct value is 0.380, and a cluster-extent p of 10^{-9} where the correct value is 0.050: inference is invalid in both directions. Reach is narrow. χ^2 statistic images are rare in applied work, the standard conjunction analysis uses the minimum statistic and is not exposed, and no high-citation empirical SPM $\{\chi^2\}$ paper was found. That is the honest reason a defect this severe ranks with,

and not above, the rest of the tier.

SP2: free energy in `spm_nlsi_GN`. The variational free energy that DCM reports and ranks models by includes a term $\frac{1}{2} \log |\Sigma^{-1}|$ in the data precision. At line 500 the precision has already been expanded across the n_q error-covariance blocks, and the code multiplies its log-determinant by n_q once more, so the term is $n_q^2/2$ times the block value rather than $n_q/2$. The M-step gradient a few lines away uses $n_y/2$, the derivative of the corrected term, so objective and gradient currently disagree. Inversions with a single block, which includes DCM for fMRI, are unaffected. DCM for evoked and induced responses, where n_q is channels times trials, has the hyperparameter-dependent part of F inflated n_q -fold, which can alter model rankings, the primary output of that method [5, 6]. This finding is confirmed by internal inconsistency and transcription; it has not been executed in MATLAB, which the pull request states.

SP3: artefact windows on epoched data. SPM’s artefact detectors write event times relative to trial onset without the peristimulus offset `timeOnset`, because the two occurrences cancel in their arithmetic. The reader `badsamples` then adds `time(this)`, which includes it. Every bad-sample window on epoched data is therefore shifted late by the baseline length: 20 samples for a 100 ms baseline at 200 Hz (Fig. 1B). Clean data are excluded and artefact partially retained, silently. The shift reaches the `mark` artefact mode, where bad channels are classified by the fraction of bad samples, the “remove bad data” option of robust averaging, cross-frequency-coupling weights and other consumers of the mask on epoched data. It does not reach the default `reject` mode, which never calls `badsamples`, nor continuous data, where the offset is zero. The machinery is that documented by the SPM8 M/EEG reference paper [8]. Merged as PR 163 and measured in Section 4.

SP4: the stamped sampling rate. `spm_eeg_downsample` computes the achieved rate and prints it, then stores the requested one. The two differ only when the user explicitly selects `decimate` or `downsample` with a non-integer ratio, because FieldTrip’s `ft_preproc_resample` then rounds the factor; the default `resample` method and the no-toolbox `fft` fallback are exact. That is narrower than the audit’s first framing, and it is recorded. Merged as PR 165 and measured in Section 4.

SP5: parametric modulators with basis-function expansion. When automatic factorial contrasts are built, `spm_design_contrasts` skips the parametric-modulation columns by inserting $\sum h$ zeros, where h counts the modulators; but every modulator column is convolved with all basis functions in `spm_Volterra`, so the design carries $\sum h \cdot n_{\text{bases}}$ such columns. Whenever temporal or dispersion derivatives are on and modulators are present, the contrast matrix is misaligned from the first modulated condition onward, and the trailing zero-pad hides the mismatch, so wrong F and T maps arrive without an error. The combination of factorial design and derivatives is the one popularised by [9]; whether specific studies also used modulators with automatic contrasts is unverified. Confirmed numerically on a constructed design; not executed in MATLAB.

3.3 The medium and low tiers

The fifteen medium-tier findings are common features with a conditional trigger, or important narrower methods. Representative examples: `spm_dicom_convert` pairs pixel spacing with the wrong direction cosines, swapping in-plane voxel sizes for non-square pixels (the spectroscopy path swaps them back with a comment saying “for some reason”); `spm_file_merge` has a sign error that clips negative intensities when merging 3-D integer volumes into 4-D; `spm_preproc_write8` returns warped and modulated tissue classes in swapped slots; `spm_dcm_peb` nests its convergence test inside `if verbose`, so with printing

off the parametric empirical Bayes routine [10] always runs 64 iterations and results depend on a display preference; `spm_kl_wishart` takes the expectation of $\log |X|$ under the wrong distribution (Monte-Carlo true divergence 6.560, coded 7.681, corrected 6.559) and `spm_lg_gamma` sums ascending arguments where the multivariate gamma descends; the windowed Fourier transform pair uses the conjugate transpose where it needs the plain one and does not round-trip; `spm_powell`'s bracket clamp has the wrong sign for descending line searches, degrading mutual-information coregistration; the FieldMap FFT smoothing grid is half a voxel off-centre on odd dimensions; `spm_robust_glm` multiplies residuals by leverage instead of dividing by $\sqrt{1-h}$, so it never down-weights anything, though only the mixture toolbox calls it; and in the active-inference engine [11] the B-novelty term indexes the action dimension with the policy index and `spm_MDP_MI`'s Dirichlet gradient is 50–100% off with the wrong sign pattern, which steers structure learning by Bayesian model reduction [12] the wrong way. Five further findings are loud crashes on documented but rarely used modes. The 63 low-tier findings are distribution edge cases, rare datatypes and formats, latent defects with no live caller, C/MEX boundary conditions, minor I/O and tabular issues, mesh and numeric utilities, M/EEG convention slips such as swapped sine and cosine phase regressors, and preprocessing minutiae, all listed with file and line in the repository.

4 Results: the two merged fixes on SPM's own data

4.1 Artefact windows: a mask that was mostly wrong, and an average that barely moved

Table 2 gives the comparison. On trial 5 the detector recorded an artefact beginning at epoch sample 30; the shipped reader marked samples 51 to 77 and the fixed reader 31 to 57 (Fig. 1B). Across the 316 channel-by-trial bad runs present under both versions the median onset shift was exactly 20 samples, the 100 ms baseline. The two masks overlapped with a Jaccard index of 0.15. Of the samples the detector had actually flagged, the shipped code excluded 24%; of the samples it excluded, 71% were clean. One hundred bad runs were truncated at the epoch end under the shipped code against 37 after the fix, so ten trials' artefacts were pushed off the edge of the epoch and lost entirely, and the total count of marked samples fell 16%, which is the quantity `mark` mode thresholds when it classifies channels as bad.

The final waveform hardly changed (Fig. 1C). The robust-averaged MMN peaked at 255 ms under both versions, with global field power 2.771 against 2.778 μV and the same largest channel at $-10.91 \mu\text{V}$; the largest difference at any channel and time was 0.37 μV , and the RMS difference over 100–300 ms was 0.044 μV against an MMN RMS of 2.94 μV , about 1.5%. The reason is that in this workflow the mask is redundant with the robust weights, which already down-weight the outlying samples the mask was meant to remove, whether or not the mask lands on them; only 84 to 94 of 600 trials carried any detected artefact, and blinks in a passive oddball are not time-locked to the deviants, so the misplaced exclusions do not bias the average systematically. Where the mask is consumed directly, in bad-channel and bad-trial classification and in the other mask consumers named above, there is no such cushion.

4.2 Sampling rate: the right number printed, the wrong one stored

Requesting 200 Hz from 512 Hz with `decimate` rounds the factor to 3, so the data are really at $512/3 = 170.667$ Hz. Both versions print “170.7 Hz”. The shipped version stores 200 Hz, so the object reports its last sample at 780.8 s of a 915.0 s recording; the fixed version stores 170.7 Hz and reports 914.8 s (Table 3). Every time value derived from the file is 14.7% too small under the shipped stamp. On the MMN this moves the global-field-power peak from its true 275.4 ms to a reported 235.0 ms (Fig. 1D), and silently mis-cuts every user-specified window: a requested -100 to 400 ms epoch covers -85 to 341 ms of real time, the baseline -85 to 0 ms, and a 100–250 ms MMN window 85–213 ms. The residual 0.02% between 170.7 and

Table 2: The artefact-window fix on SPM’s MMN tutorial recording: SPM’s own mark-mode plus robust-averaging chapter, executed with the real code in Octave, with `badsamples.m` before and after PR 163 and nothing else changed. 600 epochs of 101 samples at 200 Hz, baseline 20 samples.

	Before fix	After fix
Trial 5: detector wrote artefact at sample 30; samples marked	51–77	31–57
Median onset shift over 316 shared bad runs	20 samples (100 ms)	
Bad samples marked in total	7,584	9,017
Trials with any bad sample	84	94
Bad runs truncated at the epoch end	100	37
Jaccard overlap of the two masks	0.151	
Share of detected artefact samples actually excluded	24.2%	100%
Share of excluded samples that were clean	71.2%	0%
Robust-averaged MMN: global field power peak	255 ms (2.771 μ V)	255 ms (2.778 μ V)
Largest channel at peak (A14)	–10.91 μ V	–10.91 μ V
Largest difference, any channel and time	0.37 μ V at 300 ms	
RMS difference over 100–300 ms (MMN RMS 2.94 μ V)	0.044 μ V (1.5%)	

Table 3: The sampling-rate fix: the real `spm_eeg_downsample` in Octave, 512 Hz to a requested 200 Hz with `method = 'decimate'`, before and after PR 165. The MMN rows come from a Python transcription of the same arithmetic on the same recording.

	Before fix	After fix
Printed resampling frequency	170.7 Hz	170.7 Hz
Stored <code>fsample</code>	200.0 Hz	170.7 Hz
Reported time of last sample (true 914.994 s)	780.795 s	914.815 s
Reported MMN peak latency (true 275.4 ms)	235.0 ms	275.4 ms
Real-time extent of a requested –100 to 400 ms epoch	–85 to 341 ms	–100 to 400 ms

170.667 Hz, which the display rounding introduced and the fix inherits, is outside the fix’s scope and noted in the pull request.

5 Discussion

What the review establishes. SPM’s most travelled path is sound. The defects that were found live one step out from it, in χ^2 inference, in the M/EEG artefact and resampling machinery, in DCM for electrophysiology, in automatic contrasts with derivatives, and further out still in toolboxes and formats. Two of the five fixes at the top of the list were merged within two days of filing, each with a reproduction the maintainer could run, and three are under review; the deliverable is the merged code, and the maintainers’ reading of the rest, not mine, is what will make it trustworthy.

Severity and reach are different axes, and both were measured. The χ^2 defect is the most severe finding here, invalidating inference in both directions, and it is the one with the thinnest exposure. The artefact-window defect corrupts its mask wholesale, and on the canonical pipeline the corruption is absorbed by a second safety net. The sampling-rate defect fires only when a user picks a non-default method with a non-integer ratio, and when it fires it is not subtle. In each case the measurement narrowed the finding relative to the audit’s first framing, and the narrowing is recorded next to the finding. That is what verification is for: it moves candidates out of the confirmed set as readily as it moves them in.

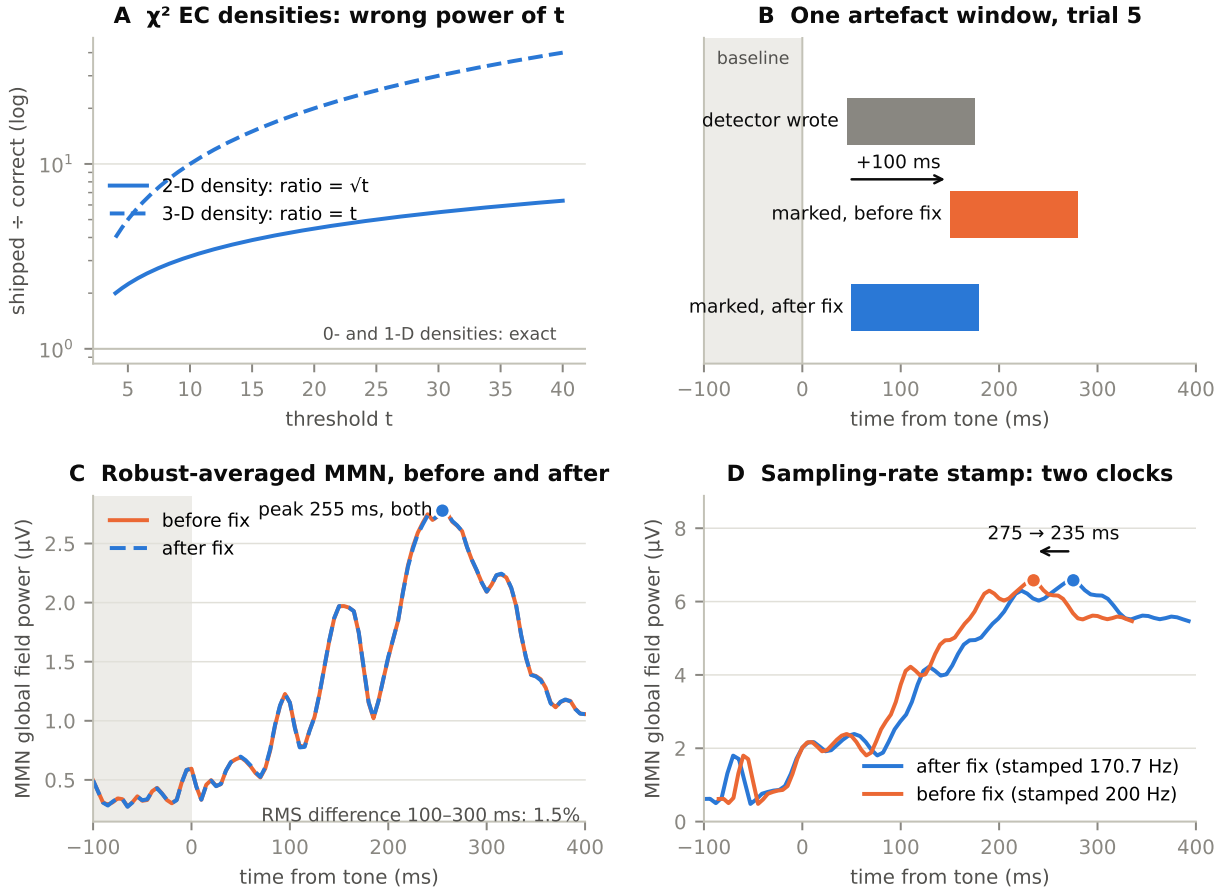


Figure 1: Two of the five high-priority defects, and the two merged ones on SPM's own data. **A**, The χ^2 Euler-characteristic densities in `spm.ECdensity`: ratio of shipped to correct density against threshold, for the 2-D ($\sqrt{t}$) and 3-D (t) orders; orders zero and one are exact. **B**, One artefact on trial 5 of the MMN tutorial recording: the window the detector recorded (grey), the samples the shipped `badsamples` marked (orange) and the samples the fixed version marks (blue). The shift is the 100 ms baseline. **C**, Global field power of the robust-averaged MMN from SPM's own artefact-removal chapter, with `badsamples` before (orange) and after (blue) the fix; the mask changed wholesale, the average by 1.5%. **D**, The MMN from the same recording decimated from 512 to a nominal 200 Hz, placed on the true time axis (blue, the rate the fixed `spm.eeg_downsample` stores) and on the axis the shipped stamp implies (orange); the true 275 ms peak is reported at 235 ms. Panels B and C use the real SPM code in Octave; panel D uses a Python transcription of the decimation arithmetic.

Why the defects survived. Each routine was correct for the data its author tested it on. The χ^2 branch had no unit test at all, and a T or F user never exercises it. The artefact detector and the artefact reader each handled the baseline consistently with themselves, and only disagreed with each other, on a path the default mode does not take. The resampler printed the right number. No user could see a defect whose signature is “artefact windows 100 ms late” or “a peak 40 ms early on a file that says 200 Hz.” This is the general shape of the findings: not carelessness, but the combinatorics of a large, old code base and evolving conventions, which no team can read exhaustively with the question “what if the offset is nonzero, or the ratio is not an integer, or there are two basis functions” in mind for every line. A machine can now ask that question of every line cheaply, and the answers arrive in the form maintainers already accept.

Limitations. “Confirmed” means traced end to end and reproduced as stated, not adjudicated by the maintainers; where they disagree, their reading wins. SPM was executed in GNU Octave, not MATLAB, with two dependencies replaced; the replaced reader was verified against an independent decoder, and the MEX files were built from SPM’s own sources, but a MATLAB-specific difference cannot be excluded. The real-data measurement is one participant and one pipeline, chosen because it is the pipeline SPM’s own documentation teaches for this recording. SP2 and SP5 have not been executed in MATLAB and their pull requests say so. The FieldMap review is partial. Exposure was assessed at the level of methods papers only; the per-study join that the AFNI companion performs for one finding has not been done for SPM, so this paper makes no claim about any individual published result. And a review of this kind is exhaustive with respect to what the agents read and checked, which is recorded, not with respect to every defect the code contains.

Data and code availability

The eleven component reviews, every verification harness (Python transcriptions, the χ^2 Monte Carlo, the Octave regression and unit-test driver), the real-data pipeline (audits/spm/reproductions/mmn_realdata/: sp3_pipeline.m, sp3_compare.py, sp4_demo.py, the Octave shims, run logs and result notes), the filing kit with issue and pull-request texts, and the script that draws Figure 1 are at <https://github.com/cindykrafft/research-software-audit>. The fixes are on <https://github.com/cindykrafft/spm> as one branch per fix, based on spm/spm at 530ec52. Upstream: issues 158, 160, 162, 164 and 167 and pull requests 159, 161, 163, 165 and 168 at <https://github.com/spm/spm>; 163 and 165 are merged. The recording is SPM’s public MMN tutorial dataset.

Disclosure of AI use

The code reviews, harnesses, patches and the Octave build described here were produced by an AI coding agent (Claude, Anthropic) operating under the author’s direction; every reported finding was verified by executing code, and every upstream filing was reviewed by the author before submission. The same system assisted in drafting this manuscript and produced its figure from the checked-in results. The author has no financial interest in any software named.

References

- [1] Friston KJ, Ashburner J, Kiebel S, Nichols T, Penny W (eds). *Statistical Parametric Mapping: The Analysis of Functional Brain Images*. Academic Press (2007).

- [2] Eklund A, Nichols TE, Knutsson H. Cluster failure: why fMRI inferences for spatial extent have inflated false-positive rates. *Proc Natl Acad Sci USA* 113:7900–7905 (2016).
- [3] Condra C. Exhaustive AI-assisted review of research software finds merged, verified defects in AFNI, and a thirteen-year-old error in regional homogeneity changes a group result. Preprint (2026); source in the same repository, [preprint/](#).
- [4] Worsley KJ. Local maxima and the expected Euler characteristic of excursion sets of χ^2 , F and t fields. *Adv Appl Probab* 26:13–42 (1994).
- [5] David O, Kiebel SJ, Harrison LM, Mattout J, Kilner JM, Friston KJ. Dynamic causal modeling of evoked responses in EEG and MEG. *NeuroImage* 30:1255–1272 (2006).
- [6] Garrido MI, Kilner JM, Kiebel SJ, Friston KJ. Evoked brain responses are generated by feedback loops. *Proc Natl Acad Sci USA* 104:20961–20966 (2007).
- [7] Näätänen R, Paavilainen P, Rinne T, Alho K. The mismatch negativity (MMN) in basic research of central auditory processing: a review. *Clin Neurophysiol* 118:2544–2590 (2007).
- [8] Litvak V, Mattout J, Kiebel S, et al. EEG and MEG data analysis in SPM8. *Comput Intell Neurosci* 2011:852961 (2011).
- [9] Henson RNA, Shallice T, Gorno-Tempini ML, Dolan RJ. Face repetition effects in implicit and explicit memory tests as measured by fMRI. *Cereb Cortex* 12:178–186 (2002).
- [10] Friston KJ, Litvak V, Oswal A, et al. Bayesian model reduction and empirical Bayes for group (DCM) studies. *NeuroImage* 128:413–431 (2016).
- [11] Friston KJ, FitzGerald T, Rigoli F, Schwartenbeck P, Pezzulo G. Active inference: a process theory. *Neural Comput* 29:1–49 (2017).
- [12] Smith R, Schwartenbeck P, Parr T, Friston KJ. An active inference approach to modeling structure learning: concept learning as an example case. *Front Comput Neurosci* 14:41 (2020).